Hall Effect Base Linear Current Sensor

Features:



- Diameter 9.0mm conductor through hole
- Output voltage proportional to AC and DC current
- Wide sensing current range 0~30 A at 5V volt.
- High sensitivity 72mV/A
- Wide operating voltage range 3.0~12 V.
- Low operating current 4.0mA
- Isolation voltage 4000V
- Ratiometric output from supply voltage
- 23K Hz Bandwidth
- Two bronze sticks for easy soldering on PCB

Functional Description:

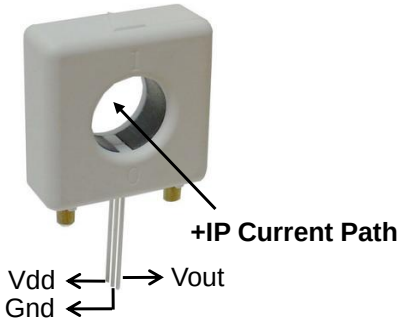
The Winson WCS1800 current sensor provides economical and precise solution for both DC and AC current sensing in industrial, commercial and communications systems. The unique package provides easy implementation without breaking original system and makes current sensing possible. Typical applications include motor control, load detection and management, over-current fault detection and any intelligent power management system etc…

The WCS1800 consists of a precise, low-temperature drift linear hall sensor IC with temperature compensation circuit and a diameter 9.0mm through hole. Users can use system's own electric wire by pass it through this hole to measure passing current. This design allows system designers to monitor any current path without breaking or changing original system layout at all. Any current flowing through this hole will generate a magnetic field which is sensed by the integrated Hall IC and converted into a proportional voltage.

The terminals of the conductive path are electrically isolated from the sensor leads. This allows the WCS1800 current sensor to be used in applications requiring electrical isolation without the use of opto-isolators or other costly isolation techniques and make system more competitive in cost.

Winson reserves the right to make changes to improve reliability or manufacturability.

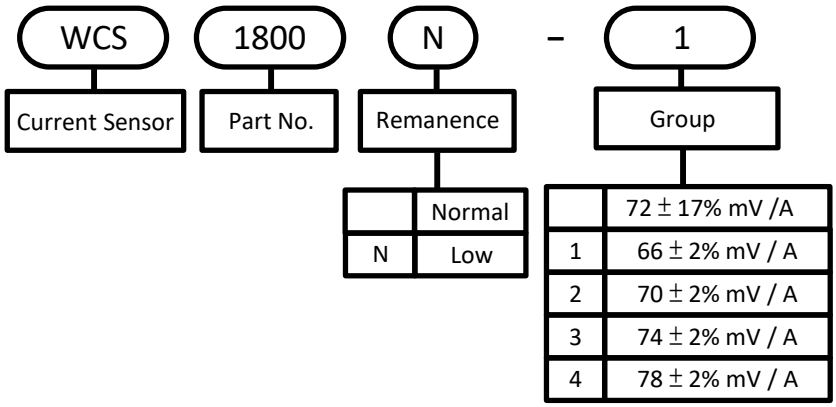
Absolute Maximum Rating (at Ta = 25˚C):



Absolute Maximum Range

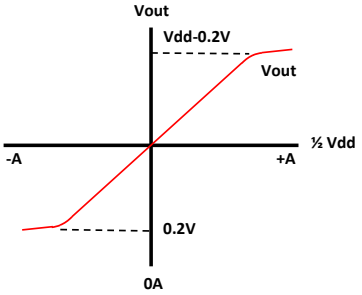
Supply Voltage, V_{DD} ---------- **14V**
Pass Through Wire Diameter ---------- **9mm**
Reverse Protection Voltage, V_R ---------- **-0.5V**
Output Sink Current, $I_{OUT(SINK)}$---------- **0.4mA**
Output Source Current, $I_{OUT(SOURCE)}$---------- **0.4mA**
Operating Temperature Range, T_A ---------- **-40ºC to +125ºC**
Storage Temperature Range, T_S ---------- **-65ºC to +150ºC**
Power Dissipation, P_D ---------- **1W**

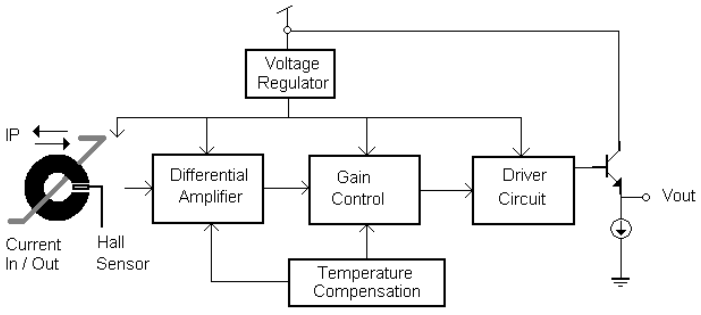
Ordering Information:



Function Block:

Vout vs. Primary Current





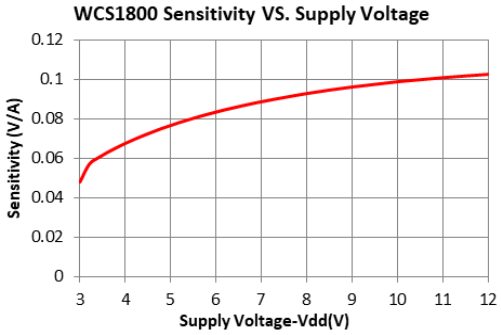
Winson reserves the right to make changes to improve reliability or manufacturability.

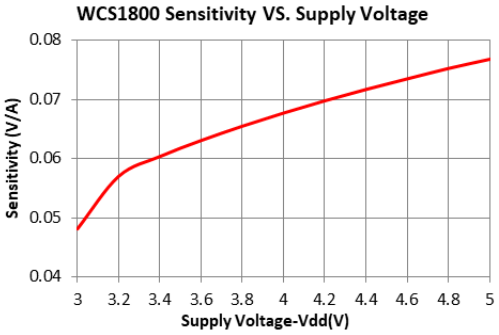
Electrical Characteristics: (T=+25˚C, Vdd=5.0V)

Characteristic	Symbol	Test Conditions		Min	Typ	Max	Units
Supply Voltage	Vdd	—		3.0	—	12	V
Supply Current	Isupply	IP =0 A		—	4.0	6.0	mA
Zero Current Vout	V0G	IP =0 A(DC Mode)		2.4	2.5	2.6	V
Conductor Through Hole	—	—		—	9.0	—	mm²
Sensitivity	Sens	WCS1800	IP= +-10 A	60	72	84	mV/A
		WCS1800-1		64.7	66	67.3	
		WCS1800-2		68.6	70	71.4	
		WCS1800-3		72.5	74	75.5	
		WCS1800-4		76.4	78	79.6	
		WCS1800N		Custom			
Bandwidth	BW	—		—	23	—	kHz
Measurable Current Range	MR	Vdd=5V (DC Mode)		—	±30	—	A
		Vdd=5V (AC RMS)		—	21	—	
Remanence Offset Error Current	I_{ERROM}	WCS1800	IP = 0 A after excursion of 30 A	—	100	—	mA
		WCS1800N		—	10	—	
Temperature Drift	△Vout	IP =0 A		—	±1.0	—	mV/℃
Output Noise	V_{Np-p}	IP =0 A		—	30	—	mV
	$V_{Np-p(0.1uF)}$	Ip =0 A, C = 0.1uF		—	4.5	—	

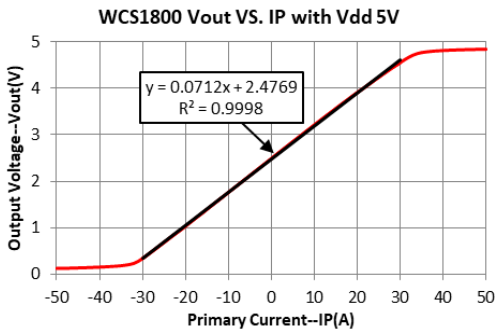
1. All output-voltage measurements are made with a voltmeter having an input impedance of at least 100kΩ

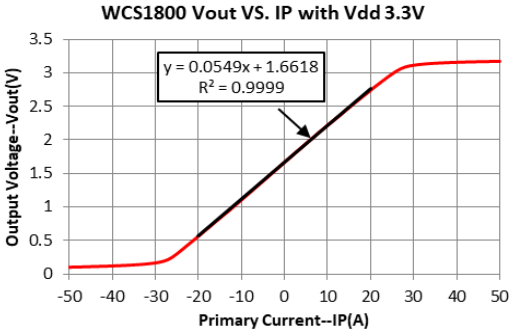
Characteristic Diagrams:

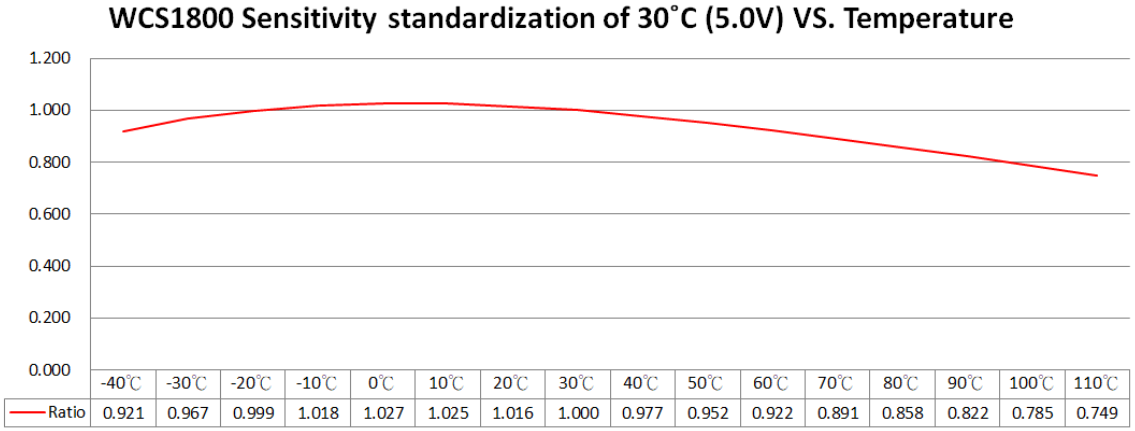




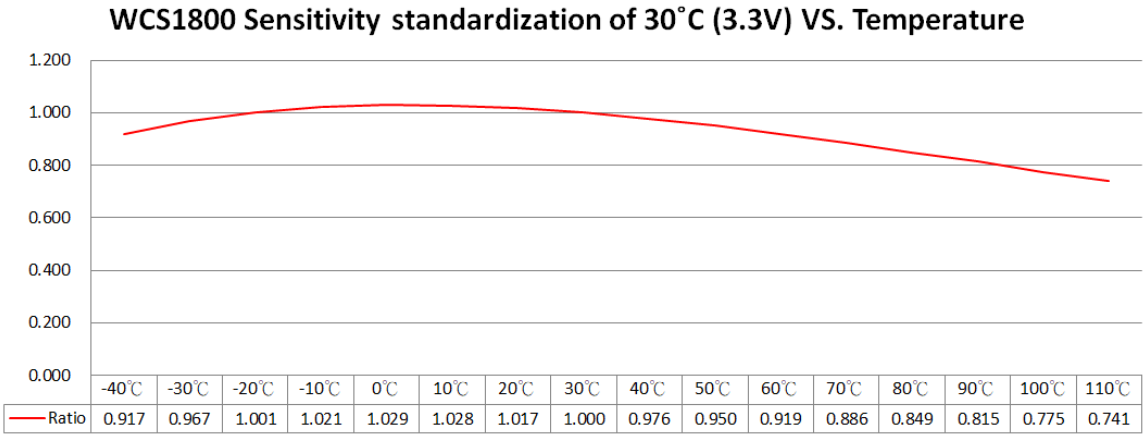
Winson reserves the right to make changes to improve reliability or manufacturability.







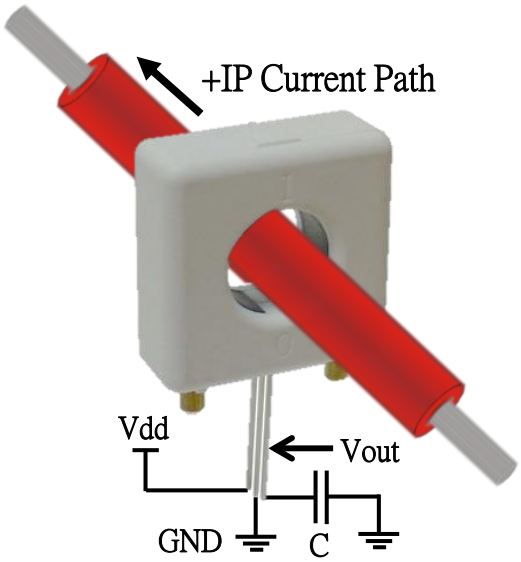
	-40℃	-30℃	-20℃	-10℃	0℃	10℃	20℃	30℃	40℃	50℃	60℃	70℃	80℃	90℃	100℃	110℃
Ratio	0.921	0.967	0.999	1.018	1.027	1.025	1.016	1.000	0.977	0.952	0.922	0.891	0.858	0.822	0.785	0.749



	-40℃	-30℃	-20℃	-10℃	0℃	10℃	20℃	30℃	40℃	50℃	60℃	70℃	80℃	90℃	100℃	110℃
Ratio	0.917	0.967	1.001	1.021	1.029	1.028	1.017	1.000	0.976	0.950	0.919	0.886	0.849	0.815	0.775	0.741

Winson reserves the right to make changes to improve reliability or manufacturability.

Application Circuit:

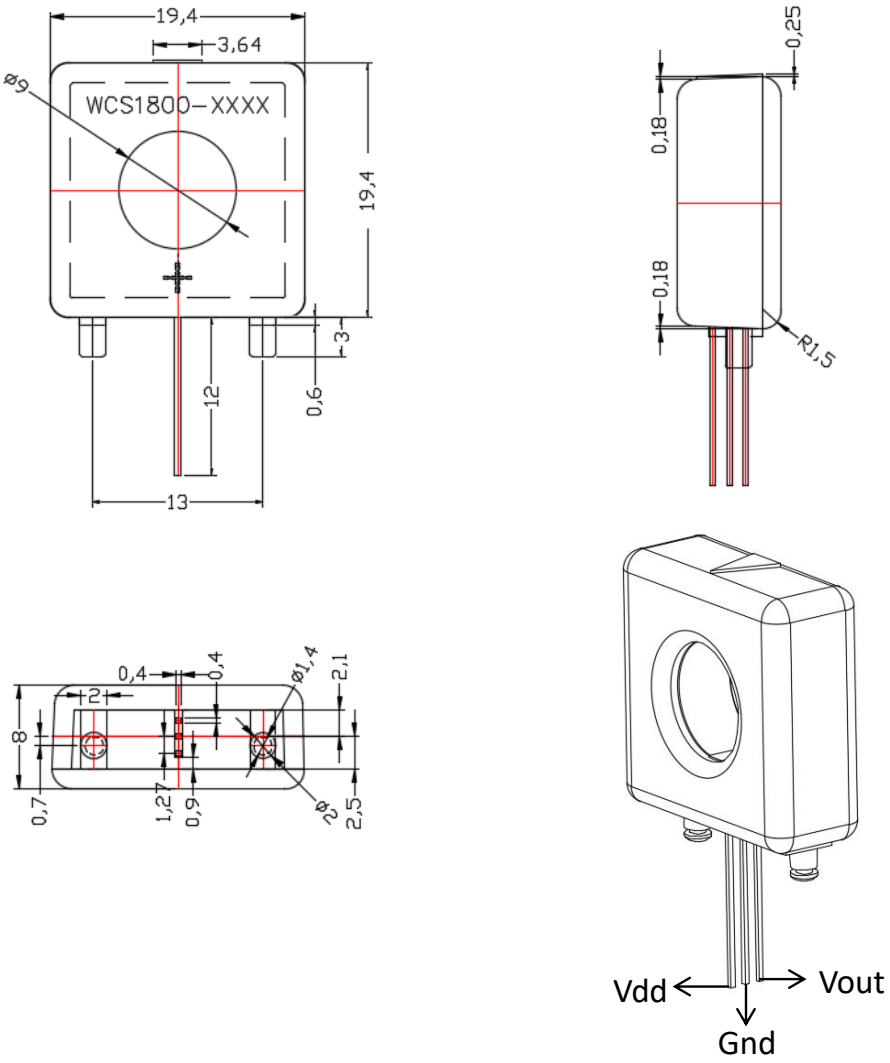


Capacitor C (0.1uF) is recommend to be connected between Vout and GND to reduce output noise.

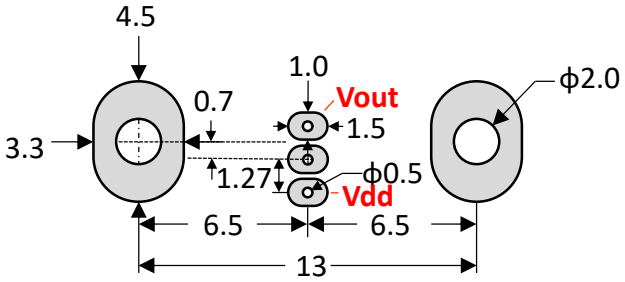
Winson reserves the right to make changes to improve reliability or manufacturability.

Package Information:

(Unit: mm)



PCB Layout Reference View (Top View)



WCS Application Note : please refer to Winson Website -> Products-> Application Note -> WCS Application Note :

http://www.winson.com.tw/Product/83

Winson reserves the right to make changes to improve reliability or manufacturability.